Q1. Check Even or Odd

```
delimiter //
create procedure check_even_odd(in num int)
begin
if Mod(num,2) = 0 then
select 'even number ' as result;
else
select 'odd number' as result;
end if;
end //
```

```
mysql> CALL check_even_odd(25);
+------------+
| result     |
+------------+
| odd number |
+------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL check_even_odd(40);
+--------------+
| result       |
+--------------+
| even number  |
+--------------+
1 row in set (0.00 sec)
```

Q2. Calculate Simple Interest

```
delimiter //
create procedure calculate_intrest(in principle int, rate int, time int)
begin
declare SI int;
set SI = principle * rate * time;
select SI as simple_intrest;
end //
```

```
mysql> call calculate_intrest(10000, 5, 2);
+----------------+
| simple_intrest |
+----------------+
|         100000 |
+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> call calculate_intrest(25000, 7, 3);
+----------------+
| simple_intrest |
+----------------+
|         525000 |
+----------------+
```

Q3. Calculate Grade

```
delimiter //
create procedure calculate_grade1(in std_id int, s_name varchar(20), s_marks int)
begin
declare s_grade varchar(10);
If s_marks is null then
select 'marks are invalid ' as message;
elseif s_marks >=90 then
set s_grade ='A+';
elseif s_marks >=80 then
set s_grade ='A';
elseif s_marks >=70 then
set s_grade ='B';
elseif s_marks >=60 then
set s_grade = 'C';
elseif s_marks >=50 then
set s_grade = 'D';
else
set s_grade ='F';
end if;
If s_marks is not null then
```

```
select
 std_id as student_id,
 s_name as student_name,
 s_marks as marks,
 s_grade as grade;
end if;
end //
```

```
mysql> CALL calculate_grade1(101, 'Amit', 85);
+------------+--------------+-------+-------+
| student_id | student_name | marks | grade |
+------------+--------------+-------+-------+
|        101 | Amit         |    85 | A     |
+------------+--------------+-------+-------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL calculate_grade1(102, 'Priya', 72);
+------------+--------------+-------+-------+
| student_id | student_name | marks | grade |
+------------+--------------+-------+-------+
|        102 | Priya        |    72 | B     |
+------------+--------------+-------+-------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_grade1(103, 'Rahul', 45);
+------------+--------------+-------+-------+
| student_id | student_name | marks | grade |
+------------+--------------+-------+-------+
|        103 | Rahul        |    45 | F     |
+------------+--------------+-------+-------+
1 row in set (0.00 sec)
```

Q4. Calculate Employee Bonus

```
delimiter //
create procedure calculate_bonus(in e_name varchar(30), e_salary decimal(10,2))
begin
declare bonus_per decimal(5,2);
declare bonus_amt decimal(10,2);
declare total_amt decimal(10,2);
```

```
case
        when e_salary >= 100000 then
                set bonus_per = 20;
        when e_salary >=50000 then
                set bonus_per = 15;
        when e_salary >=30000 then
                set bonus_per = 10;
        else
                set bonus_per = 5;
end case;
set bonus_amt = e_salary * bonus_per/100;
set total_amt = e_salary + bonus_amt;

select
        e_name as employee_name,
        e_salary as Salary,
        bonus_per as bonus_percentage,
        bonus_amt as bonus_amount,
        total_amt as total_Salary;
end //
```

----------OR----------

```
delimiter //
create procedure calculate_bonus1(in e_name varchar(30), e_salary decimal(10,2))
begin
declare bonus_per decimal(5,2);
declare bonus_amt decimal(10,2);
declare total_amt decimal(10,2);

        if  e_salary >= 100000 then
                set bonus_per = 20;
```

```
        elseif e_salary >=50000 then
                set bonus_per = 15;
        elseif e_salary >=30000 then
                set bonus_per = 10;
        else
                set bonus_per = 5;
end if;
set bonus_amt = e_salary * bonus_per/100;
set total_amt = e_salary + bonus_amt;

if e_salary is not null then
select
        e_name as employee_name,
        e_salary as Salary,
        bonus_per as bonus_percentage,
        bonus_amt as bonus_amount,
        total_amt as total_Salary;
end if;
end //
```

```
mysql> delimiter ;
mysql> CALL calculate_bonus('Amit', 120000);
+---------------+-----------+------------------+--------------+--------------+
| employee_name | Salary    | bonus_percentage | bonus_amount | total_Salary |
+---------------+-----------+------------------+--------------+--------------+
| Amit          | 120000.00 |            20.00 |     24000.00 |    144000.00 |
+---------------+-----------+------------------+--------------+--------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL calculate_bonus('Priya', 65000);
+---------------+----------+------------------+--------------+--------------+
| employee_name | Salary   | bonus_percentage | bonus_amount | total_Salary |
+---------------+----------+------------------+--------------+--------------+
| Priya         | 65000.00 |            15.00 |      9750.00 |     74750.00 |
+---------------+----------+------------------+--------------+--------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_bonus('Rahul', 25000);
+---------------+----------+------------------+--------------+--------------+
| employee_name | Salary   | bonus_percentage | bonus_amount | total_Salary |
+---------------+----------+------------------+--------------+--------------+
| Rahul         | 25000.00 |             5.00 |      1250.00 |     26250.00 |
+---------------+----------+------------------+--------------+--------------+
1 row in set (0.00 sec)
```

Q5. Find Maximum of Three Numbers —

```
delimiter //
create procedure find_maximum(in num1 int, num2 int, num3 int)
begin
declare max int default 0;
if (num1 >= num2) and (num1 >= num3) then
        set max = num1;
elseif (num2 >= num1) and (num2 >= num3) then
        set max = num2;
else
        set max = num3;
end if;
select max as maximum_number;
end //
```

```
mysql> delimiter ;
mysql> CALL find_maximum(25, 78, 45);
+----------------+
| maximum_number |
+----------------+
|             78 |
+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL find_maximum(100, 50, 75);
+----------------+
| maximum_number |
+----------------+
|            100 |
+----------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL find_maximum(30, 30, 20);
+----------------+
| maximum_number |
+----------------+
|             30 |
+----------------+
1 row in set (0.00 sec)
```

Q6. Employee Salary Analysis

```
delimiter //
create procedure salary_analysis(in e_name varchar(30), e_salary decimal(10,2))
begin
declare salary_cat varchar(20);
declare allowance decimal(10,2);
declare allow_per decimal(5,2);
declare final_salary decimal(10,2);

case
        when e_salary >= 100000 then
                set salary_cat = 'High';
                set allow_per = 20;
        when e_salary >=50000 then
                set salary_cat = 'Medium';
                set allow_per = 15;
        else
                set salary_cat = 'Low';
                set allow_per = 10;
end case;
set allowance = e_salary * allow_per/100;
set final_salary = e_salary + allowance;

select
        e_name as employee_name,
        e_salary as Salary,
        salary_cat as category,
        allow_per as allowance_percentage,
        allowance as allowance_amount,
        final_salary as final_salary;
end //
```

```
mysql> delimiter ;
mysql> CALL salary_analysis('Amit', 120000);
+---------------+-----------+----------+----------------------+------------------+--------------+
| employee_name | Salary    | category | allowance_percentage | allowance_amount | final_salary |
+---------------+-----------+----------+----------------------+------------------+--------------+
| Amit          | 120000.00 | High     |                20.00 |         24000.00 |    144000.00 |
+---------------+-----------+----------+----------------------+------------------+--------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL salary_analysis('Priya', 60000);
+---------------+----------+----------+----------------------+------------------+--------------+
| employee_name | Salary   | category | allowance_percentage | allowance_amount | final_salary |
+---------------+----------+----------+----------------------+------------------+--------------+
| Priya         | 60000.00 | Medium   |                15.00 |          9000.00 |     69000.00 |
+---------------+----------+----------+----------------------+------------------+--------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL salary_analysis('Rahul', 35000);
+---------------+----------+----------+----------------------+------------------+--------------+
| employee_name | Salary   | category | allowance_percentage | allowance_amount | final_salary |
+---------------+----------+----------+----------------------+------------------+--------------+
| Rahul         | 35000.00 | Low      |                10.00 |          3500.00 |     38500.00 |
+---------------+----------+----------+----------------------+------------------+--------------+
1 row in set (0.00 sec)
```

Q7. Electricity Bill Calculation

```
delimiter //

create procedure calculate_electricity_bill(in c_name varchar(20), unit int)

begin

declare rate int default 0;

declare total_bill int;

if unit <= 100 then

set rate = 2;

elseif unit <=200 then

set rate = 3;

elseif unit <= 400 then

set rate = 5;

else

set rate  = 7;

end if;

set total_bill = unit * rate;

select

        c_name as customer_name,
```

```
    unit as unit_consumed,
    total_bill as total_bill_amount;
end //
```

```
mysql> delimiter ;
mysql> CALL calculate_electricity_bill('Amit', 80);
+---------------+---------------+-------------------+
| customer_name | unit_consumed | total_bill_amount |
+---------------+---------------+-------------------+
| Amit          |            80 |               160 |
+---------------+---------------+-------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL calculate_electricity_bill('Priya', 150);
+---------------+---------------+-------------------+
| customer_name | unit_consumed | total_bill_amount |
+---------------+---------------+-------------------+
| Priya         |           150 |               450 |
+---------------+---------------+-------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_electricity_bill('Rahul', 350);
+---------------+---------------+-------------------+
| customer_name | unit_consumed | total_bill_amount |
+---------------+---------------+-------------------+
| Rahul         |           350 |              1750 |
+---------------+---------------+-------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL calculate_electricity_bill('Sneha', 500);
+---------------+---------------+-------------------+
| customer_name | unit_consumed | total_bill_amount |
+---------------+---------------+-------------------+
| Sneha         |           500 |              3500 |
+---------------+---------------+-------------------+
1 row in set (0.00 sec)
```